MILAN SINGH

FULL-STACK & REACT NATIVE DEVELOPER
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SUMMARY

Full-Stack and React Native Developer who designs, builds and ships production apps end-to-end. Built and published a **full-featured mobile app on the Google Play Store** with OTP authentication, push notifications, real-time updates and device connectivity, backed by **Fastify, PostgreSQL, Redis and AWS**.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, SQL
Mobile: React Native (Expo), Play Store Release, BLE, Wi-Fi (Local AP), OTA Updates, Push Notifications
Frontend: React, Next.js, Redux Toolkit, Zustand, TanStack Query, React Hook Form, Zod, Tailwind CSS
Backend: Node.js, Fastify, Express.js, REST APIs, WebSockets, MQTT, BullMQ, JWT, Clerk, Better Auth
Databases: PostgreSQL, MongoDB, Redis, Appwrite
Cloud & DevOps: AWS (Lambda, EC2, S3, API Gateway, Aurora, CloudFront), Docker, GitHub Actions, Jest
Concepts: System Design, Microservices, Caching, Rate Limiting, RBAC, AI Integration (Vercel AI SDK)

PROFESSIONAL EXPERIENCE

Software Developer, Dash Dynamic | IIT Delhi **Sep 2025 – Present**

- Built and published the company's production **React Native (Expo)** app on the **Google Play Store**, covering the full user flow from sign-in to real-time device control.
- Implemented **OTP authentication** with SMS auto-read and **rate limiting**, and a **push notification** system using BullMQ queues, chunked sends and dead-token pruning.
- Connected the app to ESP32 devices over **BLE** and the device's **local Wi-Fi access point**, with auto-reconnection, failover and offline support.
- Built live device and system updates using **WebSockets** across the mobile app and dashboard.
- Delivered an end-to-end **OTA firmware update** pipeline with SHA-256 verification and controlled releases.
- Designed **Fastify** services on **PostgreSQL** with **Redis** caching, and **MQTT** telemetry pipelines for continuous charger availability with load balancing and failover.
- Built an operations dashboard with real-time charts, map routing and **role-based access control**, on **AWS**.

Frontend Developer, Sudo Yantra | New Delhi **Nov 2024 – Feb 2025**

- Reduced initial load time through **code splitting**, lazy loading and asset optimization in an Agile team.
- Raised accessibility to **WCAG AA** and cut image bandwidth by **~40%** with WebP and responsive images, improving Core Web Vitals.

PROJECTS

BingeBee – Video Streaming PlatformLive | GitHub

React, TypeScript, Express.js, MongoDB, Redis, Cloudinary, TanStack Query, Vercel

- Built a full-stack **YouTube-style** video platform where visitors can browse videos publicly, and signed-in users can upload, watch and manage their own profile.
- Implemented video and media upload using **Multer** and **Cloudinary**, with uploaded files stored and served from the cloud.
- Built social features: **subscriptions, likes, comments, playlists, watch history** and **community posts**.
- Wrote a custom **Express.js** REST backend with **JWT** auth stored in **httpOnly cookies** for secure sessions.
- Designed **MongoDB aggregation pipelines** to generate the video feed, subscription and subscriber lists, and watch history.

- Built a type-safe **React + TypeScript** frontend with **TanStack Query & Router**, React Hook Form and Zod, using client-side caching for fast navigation.

Happy Dates – Birthday Reminder App | Android & iOS

React Native (Expo), TypeScript, Vercel AI SDK, Better Auth, TanStack Query, Zustand, Express.js, FCM

- Built a cross-platform **React Native (Expo)** app to save birthdays and get timely reminders, with onboarding, a searchable month-wise list, add / edit / delete with photo upload, profile management and light / dark themes.
- Built an **AI assistant** with the **Vercel AI SDK** that lets users chat to plan events, generate personalized birthday wishes and messages, and mark important dates on the calendar.
- Implemented **push notifications** with Expo Notifications and **Firebase Cloud Messaging (FCM)**: Android notification channel setup, device token registration with the backend, and reminders 7 days before, 3 days before and on the morning of each birthday.
- Added email / password authentication with **Better Auth**, sessions stored securely on the device, and protected routes with **Expo Router** guards so signed-out users cannot reach app screens.
- Built the data layer with **TanStack Query** caching and mutations, **Zustand** for app state, an Axios client with auth interceptors, **Zod**-validated forms and offline / online status banners.

ACHIEVEMENTS

Smart India Hackathon 2022 – First Runner-Up **2022**

- Built a real-time dashboard with sensor integration for driver-health monitoring to help prevent heavy-vehicle accidents.

EDUCATION

B.Sc. in Electronic Science – University of Delhi **Jul 2021 – Jun 2024**